OCT changes in peri-tumour normal retina following ruthenium-106 and proton beam radiotherapy for uveal melanoma

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ABSTRACT

Introduction Uveal melanoma is most commonly treated with radiotherapy, destroying the tumour cells with adequate safety margins and limiting collateral damage to surrounding structures to preserve maximal vision. We used optical coherence tomography (OCT) to study the effects of radiotherapy on the retina.
Methods Patients with posteriorly located choroidal melanoma treated with proton beam radiotherapy (PBR) and ruthenium-106 brachytherapy between January 2010 and June 2014 underwent spectral domain OCT.
Results Images of 32 patients following ruthenium-106 brachytherapy and 44 patients following proton beam teletherapy were analysed. Following plaque brachytherapy, an early marked disruption of the outer retinal layers could be observed in 30 cases (94%) with retinal atrophy evident in 26 cases (81%). In contrast, the images from patients who underwent PBR showed subtle outer retinal layer change with 16 cases (36%) showing some inner-outer segment junction disruption by 6 months and 63% by 24 months with minimal atrophy. In cases with tumours <2 mm from the fovea, the visual loss was significantly less at 6 and 12 months in the proton beam group.
Conclusion In comparison to ruthenium-106 plaque brachytherapy, PBR leads to more subtle and slower changes in the outer retinal layers enabling retention of visual function for longer. The difference in dosing regime and dose distribution across the tumour is likely to be causative for this structural differential.

INTRODUCTION

Brachytherapy and teletherapy have been used in the treatment for choroidal melanoma for many decades.[1–3] Radiotherapy of choroidal melanoma near the fovea and optic disc poses particular challenges. The aims of the treatment are to destroy all tumour cells with an adequate margin of safety and, at the same time, limit the damage to the surrounding structures in order to preserve as much vision as possible. Over the past decades, various treatment strategies have been developed in order to minimise the damage to the tissue surrounding the tumour. These include eccentric placement of ruthenium-106 plaques[4] and proton beam radiotherapy (PBR), which use a highly collimated treatment beam.[5] Eccentric placement of plaques involves covering the base of the tumour with the plaque in a non-central fashion, reducing the safety margins towards the fovea to preserve macula function.

Spectral domain and swept source optical coherence tomography (OCT) techniques enable the accurate identification of retinal layers including ganglion cell layer, inner plexiform layer (IPL), inner nuclear layer (INL), outer plexiform layer (OPL), outer nuclear layer (ONL), outer limiting membrane, inner segments, outer segments, retinal pigmented epithelium and Bruch's membrane which have been correlated to histologic layers in rat and macaque models.[6–8] In addition, the choriocapillaris and choroid can now be routinely visualised.[9] Making use of these improvements, OCT has been increasingly employed in the oncology field, with characteristic findings indicative of exudative activity (such as the presence of subretinal fluid) or chronicity (such as overlying retinal atrophy) when differentiating between benign and malignant choroidal lesions.[10] OCT is also used to detect and monitor the well-documented risk of development of radiation maculopathy following plaque or PBR[11 12] with the development of intraretinal/subretinal fluid.

The aim of this study was to characterise the OCT findings of normal peri-tumour tissue following ruthenium-106 plaque brachytherapy and PBR for choroidal melanoma. This may aid in modifying treatment parameters for radiotherapy of posterior pole melanomas with the aim to further reduce the damage to the structures surrounding the tumour. Primary outcome measures were visible changes in retinal layers demonstrable on OCT imaging and secondary measures were impacts on visual acuity outcomes.

METHODS

We performed a retrospective review of patients treated with ruthenium-106 brachytherapy and PBR at the Liverpool Ocular Oncology Centre between January 2010 and June 2014. The surgical details of treatment have previously been published elsewhere[4 5]; however, a brief description follows.

Ruthenium-106 plaque placement

To ensure accurate positioning of a plaque in relation to the tumour, any overlying extraocular muscle was routinely disinserted. The anterior and lateral tumour margins were identified by transpupillary transillumination or indentation. Tumours distant to fovea were treated with a minimum safety margin of 2 mm; however, those very close to the fovea were treated with the posterior edge of the plaque aligned to the posterior tumour margin.

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Correct positioning was confirmed by inserting the tip of a right-angled, 20-G transilluminator behind the plaque and performing binocular indirect ophthalmoscopy. The plaque was considered to be in good position if the light from the transilluminator was located at the posterior tumour margin (ie, 'sunset sign'). Any disinserted muscles were returned to their correct anatomic position. The plaque remained in situ until a scleral dose of 350 Gy with a minimal tumour apical dose of 80–90 Gy was administered, according to calculations predicted by radiation physicists at the Clatterbridge Oncology Centre.

Tantalum marker insertion

In preparation for PBR, tantalum marker insertion was undertaken. A 180° conjunctival peritomy was performed and recti muscles were slung with 2-0 silk. The tumour margins were identified by transpupillary transillumination, or if this was not possible, by binocular indirect ophthalmoscopy with indentation. Four tantalum markers were sutured to the sclera with 5-0 mersilene sutures. They were positioned as close as possible to the anterior, lateral and posterior tumour margins, preferably over normal choroid. Measurements were taken between each marker, to the limbus and to the tumour edge. The patients were simulated and treated at the Clatterbridge Douglas Cyclotron. A face mask and dental bite were prepared for each patient to immobilise the head precisely. The position and shape of both the eye and tumour were modelled, and the optimum gaze angle was finalised. The required range and modulation for each treatment were calculated, with standard lateral and distal safety margins of 2.5 mm. Four daily fractions were administered, on consecutive days, with a total dose of 53.1 proton Gy.

Since January 2010, all patients undergoing radiotherapy for choroidal melanoma underwent routine examination with OCT analysis of the posterior pole using the Spectralis OCT (Heidelberg Engineering, Heidelberg, Germany) with 19-line volume scan with a setting of 9 ART (automatic real time) over the fovea. All treated melanomas were identified but only posterior pole tumours were chosen for analysis to ensure that there was adequate quality of OCT scans through both the tumour and radiation field; only patients who were undergoing primary treatment were included. Normal sections of retina were analysed >200 µm away from the visualised tumour margin with the same single horizontal section assessed for each visit; if available for consistency, this was located at the fovea (this was not possible for nasal located tumours). All images were viewed and graded by two independent assessors (RH and FH) following a standardised protocol; each retinal layer at each time point was graded as normal, disrupted or absent. If the assessor gradings differed, a third opinion was sought (HH).

Informed patient consent was obtained for surgical procedures as well as collection and analysis of all data. Ethics committee approval was not required for this retrospective data analysis. The study adheres to the tenants of the Declaration of Helsinki. Statistical methods included a one-way analysis of variance (ANOVA) to assess for differences between the two groups of data.

RESULTS

Ruthenium-106 plaque brachytherapy

Between the chosen time frame, 201 patients underwent ruthenium-106 plaque brachytherapy for choroidal melanoma. The majority of these were peripheral or mid-peripheral lesions. Adequate OCT frames of posterior tumours prior to and following plaque radiotherapy were obtained for 32 patients. There was a men:women ratio of 1:1 (16 men and women); the mean age at treatment was 59 years (range 19–86 years). All had at least 24-month follow-up.

The mean longest basal diameter was 9.02 mm (range 4.20–14.6 mm, SD 3.06) and mean thickness 2.50 mm (range 1.52–5.7 mm, SD 1.34). The mean distance to the fovea was 3.92 mm (SD 3.30) and distance to the disc was 5.44 (SD 2.67) (table 1).

Preoperatively, OCT sections of peri-tumour retinal tissue demonstrated subretinal fluid (n=3), hyper-reflective foci (HRF) (n=10), partial disruption of the ELM (n=9) and loss of the inner–outer segment junction (IS/OS) (n=2) (both had associated subretinal fluid) with associated retinal atrophy (n=1) (table 2).

Preoperatively, OCT sections of peri-tumour retinal tissue demonstrated subretinal fluid (n=17), HRF (n=4) and loss of the IS/OS (n=7) with no associated retinal atrophy (table 3).

At 6-month follow-up, 2 patients developed epiretinal membranes, 2 had intraretinal fluid, 1 had subretinal fluid and 17 developed HRF. The ELM and IS/OS layer were disrupted in 30 patients. Twenty-six patients developed marked retinal atrophy over this time period (figure 1).

Table 1 Comparison of proton beam radiotherapy (PBR) and plaque groups.

	Ruthenium-106	PBR	P value
Age (years)	59 (19–86)	58 (40–84)	0.68
Gender (M:F)	16:16	29:15	
Diameter (mm)	9.02 (4.2–14.6)	9.6 (5.72–15.44)	0.48
Thickness (mm)	2.50 (1.52–5.7)	2.9 (1.47–6.57)	0.10
Distance to fovea (mm)	3.92 (0–10.0)	2.0 (0–8.0)	0.02*
Distance to disc (mm)	5.44 (0–12.0)	3.6 (0–8.0)	0.008*
Scleral dose(Gy)	374.1 (297–642)	53.1	<0.001*
Apical dose (Gy)	158.6 (74–214)	53.1	<0.001*

Baseline characteristics show similar patient age demographics and tumour dimensions but the PBR group tend to have tumours closer to the fovea and optic disc. The dosage applied to the ocular structures is much higher in the plaque group. *show results of significance with p<0.05

Table 2 Number of patients in which abnormalities were demonstrable on optical coherence tomography before and following ruthenium-106 plaque brachytherapy including presence of epiretinal membrane (ERM), intraretinal fluid (IRF), subretinal fluid (SRF), hyper-reflective foci (HRF) and disruption of the inner plexiform layer (IPL), inner nuclear layer (INL), outer plexiform layer (OPL), outer nuclear layer (ONL), external limiting membrane (ELM), inner–outer segment junction (IS/OS) and retinal pigment epithelium (RPE), as well as complete retinal atrophy

Plaque (n=32)	Baseline	6 months	12 months	24 months
ERM	0	2	2	3
IRF	0	2	0	0
SRF	3	1	0	0
HRF	10	17	4	4
IPL	1	27	30	32
INL	1	27	30	32
OPL	1	29	31	32
ONL	1	29	31	32
ELM	9	30	32	32
IS/OS	2	30	32	32
RPE	0	0	0	1
Atrophy	1	26	31	32

Table 3 Number of patients in which abnormalities were demonstrable on optical coherence tomography before and following proton beam radiotherapy (PBR) including presence of epiretinal membrane (ERM), intraretinal fluid (IRF), subretinal fluid (SRF), hyper-reflective foci (HRF) and destruction of the inner plexiform layer (IPL), inner nuclear layer (INL), outer plexiform layer (OPL), outer nuclear layer (ONL), external limiting membrane (ELM), inner–outer segment junction (IS/OS) and retinal pigment epithelium (RPE), as well as complete retinal atrophy

PBR (n=44)	Baseline	6 months	12 months	24 months
ERM	0	1	1	1
IRF	1	1	3	10
SRF	17	1	5	6
HRF	4	6	6	24
IPL	0	1	1	6
INL	0	1	0	6
OPL	0	1	0	6
ONL	0	10	4	8
ELM	1	5	5	7
IS/OS	7	16	18	28
RPE	0	2	2	4
Atrophy	0	1	1	2

Figure 1 Optical coherence tomography of foveal area prior to and following eccentric ruthenium-106 plaque brachytherapy at 6, 18 and 24 months.

At 12 months, there were still 2 cases of epiretinal membrane, no cases of intraretinal/subretinal fluid and 4 developed HRF; all cases had disruption of the ELM and IS/OS layer with 31 having marked retinal atrophy. By this period, these changes demonstrated a sharp demarcation between normal and treated retina.

At 24-month follow-up, all cases demonstrated total ELM and IS/OS loss with retinal atrophy, and there were no further changes in the appearance of the OCT compared to that at 12 months (figure 2).

Visual acuity in this group was mean logMAR 0.17 (range −0.1 to 0.4) Snellen 6.7.5 (range 6/5–6/15) at baseline which dropped to 0.32 (Snellen 6/12) at 6 months and stable at 0.32 (Snellen 6/12) at 12 months and beyond. In the subset of patients (n=12) with lesions within 2 mm of the fovea, the baseline acuity was mean 0.3 logMAR (Snellen 6/12) which dropped to 0.8 (Snellen 6/36) at 6 months and 0.9 (Snellen 6/48) at 12 months.

Proton beam radiotherapy

Between the chose time frame, 223 patients underwent PBR for choroidal melanoma. Many were ciliary body lesions or diffuse lesions with unclear margins. Adequate OCT frames of posterior tumours prior to and following PBR were obtained for 44 patients. There was a men:women ratio of 2:1 (29 men 15 women); the mean age at treatment was 58-year range (32–84 years).

The mean longest basal diameter was 9.60 mm (range 5.72–15.44 mm, SD 3.08) and mean thickness was 2.90 mm (range 1.47–6.57 mm, SD 1.62). The mean distance to the fovea was 2.00 mm (SD 3.40) and distance to the disc was 3.60 (SD 2.88) (table 1).

At 6-month follow-up, 1 patient developed epiretinal membrane, 1 intraretinal fluid and 6 HRF. The ELM was disrupted in 5 patients, with IS/OS disruption in 16. A single patient developed retinal atrophy.

At 12 months, there was a single case of epiretinal membrane, 3 had intraretinal and 5 had subretinal fluid and 6 had HRF. ELM was disrupted in 5, IS/OS in 18 but retinal atrophy in just 1 case (figure 2).

At 24-month follow-up, 16 patients demonstrated intraretinal or subretinal fluid due to radiation retinopathy, 24 developed HRF foci, 7 patients with ELM disruption, 28 with IS/OS disruption and only 2 cases of retinal atrophy (figure 3).

Visual acuity in the PBR group was mean logMAR 0.1 (range −0.2 to 0.5) Snellen 6/7.5 (range 6/4–6/18) at baseline which dropped to 0.2 (Snellen 6/9) at 6 months, 0.3 (Snellen 6/12) at 12 months and 0.5 (Snellen 6/18) at 18 and 24 months. In the subset of patients (n=31) with lesions within 2 mm of the fovea, the baseline acuity was mean 0.1 logMAR (Snellen 6/7.5) which dropped to 0.2 (Snellen 6/9) at 6 and 12 months and 0.4 (Snellen 6/15) at 18 and 24 months post-treatment (figure 4).

The baseline patient and tumour characteristics were similar in the two groups (table 1); however, the PBR tumours were generally closer to the fovea and the optic disc than those treated with plaque brachytherapy ($p=0.02$ and $p=0.008$, respectively). There was a significant difference in dose applied to the ocular structures with plaque treatments receiving much higher scleral and apical tumour doses ($p<0.001$). Statistical analysis (one-way ANOVA) demonstrated significant differences in the rate of development in HRF following PBR ($p<0.05$) in keeping with the time points for expected radiation maculopathy (24 months). There were significant differences at all post-treatment time points in the rate of disruption and destruction of the IPL, INL, OPL, ONL and the development of retinal atrophy ($p<0.001$) with plaque treatments in comparison to PBR. Disruption of the IS/OS junction was significantly different at 6 and 12 months but not so at 24 months ($p=0.4$)

The visual acuity differences between the two groups in total are not significant ($p=0.6$) at 6 or 12 months (one-way ANOVA); however, for those cases with lesions within 2 mm of the fovea, the difference in vision becomes significant at 6 months post-treatment ($p=0.01$), and at 12 months ($p=0.005$) with the proton beam-treated patients retaining better visual acuity than the plaque brachytherapy patients. This was not continued at 18 months ($p=0.3$) and beyond (figure 4).

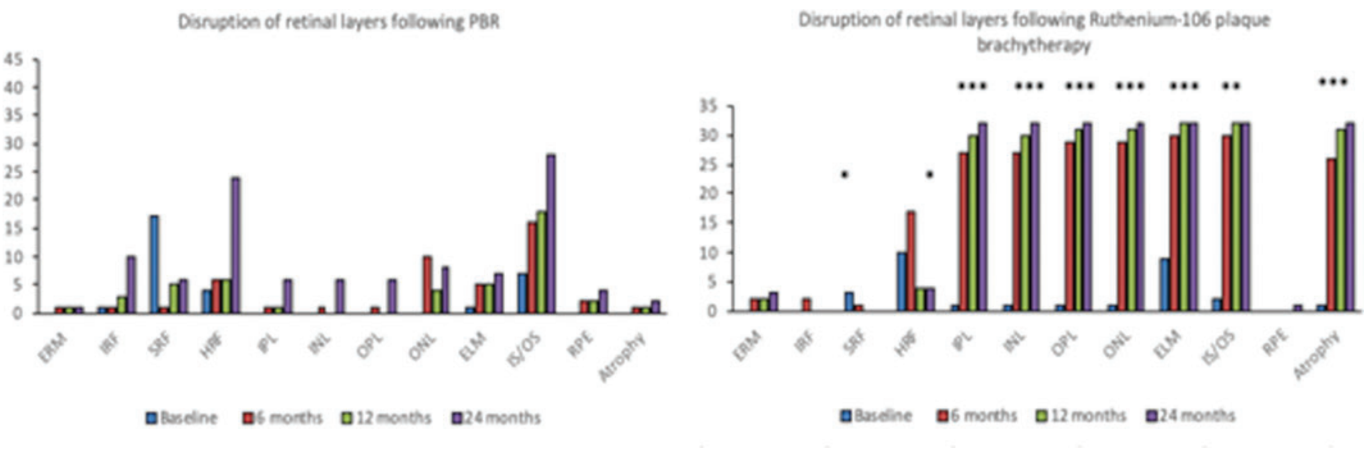

Figure 2 Graphical representation of number of patients with optical coherence tomography changes over time in both the proton beam radiotherapy (PBR) and plaque groups. Statistical significance of difference between the groups is annotated with *p<0.05, **p<0.01, ***p<0.001. ELM, external limiting membrane; ERM, epiretinal membrane; HRF, hyper-reflective foci; INL, inner nuclear layer; IPL, inner plexiform layer; IRF, intraretinal fluid; IS/OS, inner–outer segment junction; ONL, outer nuclear layer; OPL, outer plexiform layer; RPE, retinal pigment epithelium; SRF, subretinal fluid disruption.

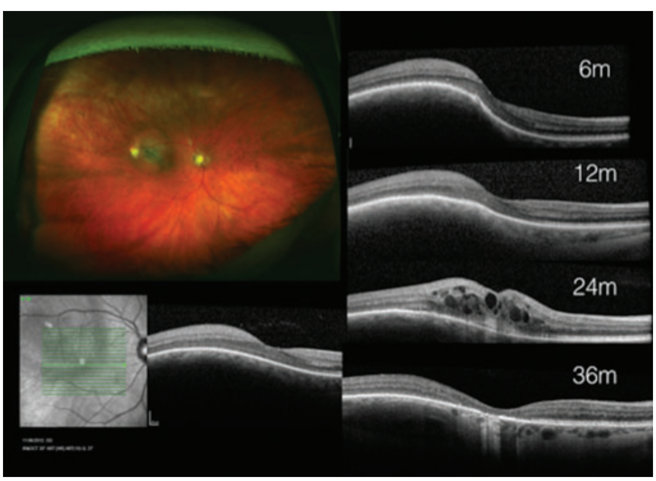

Figure 3 Optical coherence tomography of foveal area prior to and following proton beam radiotherapy at 6, 18, 24 and 36 months.

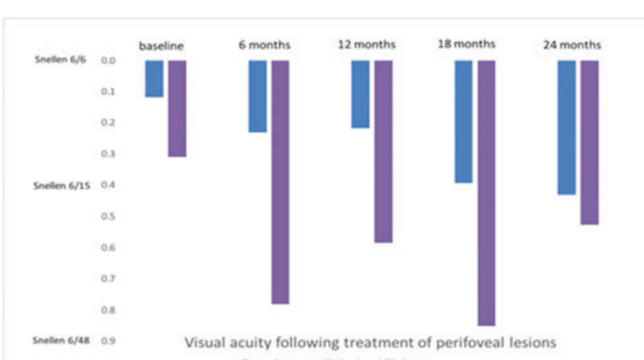

Figure 4 Mean visual acuity of patients treated for lesions <2 mm from the fovea following ruthenium-106 plaque brachytherapy and proton beam radiotherapy.

DISCUSSION

OCT has been used within the ocular oncology sector for the detection of suspicious features of choroidal lesions[10] as well as measuring smaller lesions with the benefit of EDI (enhanced depth imaging)-OCT.[11] Although OCT has been used to monitor patients treated for choroidal melanoma with radiotherapy, the emphasis has been to analyse the resolution of subretinal fluid following regression of the tumour, as well as detecting the development of longer-term complications such as radiation retinopathy.[12 13]

The results in the ruthenium-106 brachytherapy group demonstrate a rapid progression of outer retinal damage involving the ELM and IS/OS layers over the course of 6–12 months. By the end of the 12-month period, there was marked outer retinal layer disruption and loss with global retinal atrophy in the majority of patients in collateral peri-tumour radiation-treated tissue; this was noted to have a fairly abrupt demarcation with normal untreated areas of the retina which retained normal stratum structure (the layered configuration of the retina was preserved). In contrast, the group treated with PBR showed a much more subtle change in retinal structure over the same time period with very little destructive or atrophic changes even after 24 months. Although patients treated with PBR generally had tumours closer to the fovea and optic disc, likely as these lesions are more difficult to treat adequately with plaque therapy and therefore were surgeon selected to undergo PBR, this differential in OCT changes may explain the retention of better visual acuity in the short term and in the medium term in patients treated with PBR for lesions close to the fovea.

The dosing regime of PBR versus ruthenium-106 plaque brachytherapy is very different. Plaque placement involves a much higher scleral level dose than apical tumour dose due to the dose differential over distance and the reduction in radiation dose with increasing penetration through tissue. The Liverpool dosing has a minimal scleral dose of 350 Gy to enable a plaque footprint which, especially with small tumours, administers a much higher dose to the apex and adjacent tissues than required. PBR in contrast enables an even dose across the whole tumour due to the nature of the bragg peak, enabling a consistent dose of 53.1 Gy to the tumour and adjacent tissues within the safety margin. This significant dose differential will of course be largely contributory to the OCT changes demonstrated here. Although it would be interesting to analyse equivalent dosing with these two radiation sources, this report aims to give real-world data to analyse real-life impact on patients and outcomes. Further analysis on ex vivo models may give us more understanding on the differences in equivalent dose tissue damage; however, with this further understanding of the time course of the destructive qualities of these standard dose therapies on normal peri-lesional tissue, we may be able to counsel patients more accurately as to expected visual outcomes of centrally treated choroidal tumours.

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REFERENCES

1 Stallard HB. Radiotherapy of malignant intra-ocular neoplasms. *Br J Ophthalmol* 1948;32:618–39.
2 Stallard HB. Radiotherapy for malignant melanoma of the choroid. *Br J Ophthalmol* 1966;50:147–55.
3 Gragoudas ES, Goitein M, Koehler AM, *et al.* Proton irradiation of small choroidal malignant melanomas. *Am J Ophthalmol* 1977;83:665–73.
4 Russo A, Laguardia M, Damato B. Eccentric ruthenium plaque radiotherapy of posterior choroidal melanoma. *Graefes Arch Clin Exp Ophthalmol* 2012;250:1533–40.
5 Damato B, Kacperek A, Chopra M, *et al.* Proton beam radiotherapy of choroidal melanoma: the Liverpool-Clatterbridge experience. *Int J Radiat Oncol Biol Phys* 2005;62:1405–11.
6 Drexler W, Morgner U, Ghanta RK, *et al.* Ultrahigh-resolution ophthalmic optical coherence tomography. *Nat Med* 2001;7:502–7.
7 Berger A, Cavallero S, Dominguez E, *et al.* Spectral-domain optical coherence tomography of the rodent eye: highlighting layers of the outer retina using signal averaging and comparison with histology. *PLoS One* 2014;9:e96494.
8 Toth CA, Narayan DG, Boppart SA, *et al.* A comparison of retinal morphology viewed by optical coherence tomography and by light microscopy. *Arch Ophthalmol* 1997;115:1425–8.
9 Laviers H, Zambarakji H. Enhanced depth imaging-OCT of the choroid: a review of the current literature. *Graefes Arch Clin Exp Ophthalmol* 2014;252:1871–83.
10 Say EA, Shah SU, Ferenczy S, *et al.* Optical coherence tomography of retinal and choroidal tumors. *J Ophthalmol* 2012.
11 Shields CL, Manalac J, Das C, *et al.*, Review of spectral domain enhanced depth imaging optical coherence tomography of tumors of the choroid. *Indian J Ophthalmol* 2015;63:117–21.
12 Horgan N, Shields CL, Mashayekhi A, *et al.* Early macular morphological changes following plaque radiotherapy for uveal melanoma. *Retina* 2008;28:263–73.
13 Shah NV, Houston SK, Markoe AM, *et al.* Early SD-OCT diagnosis followed by prompt treatment of radiation maculopathy using intravitreal bevacizumab maintains functional visual acuity. *Clin Ophthalmol* 2012;6:1739–48.